

Schema Design for E-Commerce DB

Customer (CustomerId PK, FirstName, LastName, Email, Phone, Address)

Staff (StaffId PK, FirstName, LastName, Email, Role, DateHired)

CreditCard (CardId PK, CustomerId FK (references Customer(CustomerId)), CardNumber, CardHolder, DateExpires, ZipCode)

Product (ProductId PK, Name, Description, Category, Price, Stock, ManagedBy FK (references Staff(StaffId)))

Purchase (PurchaseId PK, CustomerId FK (references Customer(CustomerId)), ProductId FK (references Product(ProductId)), CardId FK (references Card(CardId)), Quantity, TotalPrice, DatePurchased)

Relationships Table		
Relationships	Cardinality	Notes
Staff – Manages - Product	1:N	Staff can add and edit products
Customer – Owns – Credit Card	1:N	Customers can enter multiple cards into the e-commerce site
Customer – Makes - Purchase	1:N	Customers can make and look at prior purchases
Product – Is - Purchased	1:N	Products can be in multiple purchases until stock empties
Credit Card – Pays for - Purchase	1:N	Every purchase has to be made with a credit card